3 Part 3 – Further Study

3.1 Assessment Requirements – Further Study

For part 3 you will be assigned a theme from one of the following:

1. Machines
2. Converters

Using the **case study parameters from weeks 1 to 7** (not the ones you have been assigned for the assessment) and the analyses/simulations you have developed during the teaching in weeks 1 to 7, undertake a design study that explores an aspect of your assigned theme in more detail.

Alternatively, you could use the theory presented in the Switches lectures in week 7 to develop an analysis and simulation to determine the loss in the inverter switches over a single AC cycle at the case study maximum speed.

Imagine that you are attempting to convince the technical director of your future company of the suitability (or not) of an idea that you have had or scenario you want to explore prior to them making a significant purchasing design in your assigned theme.

You must come up with an idea (or base yours on one of those listed below) to explore with a reason and then undertake your design space exploration. You need to show (and document) that you can

1. Come up with and elaborate an idea that you can explore in your assigned theme area using the design tools you have made in weeks 1 to 7. The more context aware your study is relative to the case study, the better the potential outcome. You are trying to demonstrate how you would use the tools you have developed to explore ideas related to the case study and recommend changes based upon your results.
2. Determine how you can use the design tools to undertake the exploration or design study you have identified in step 1.
3. Use the simulation tools to undertake the study you have outlined in step 2.
4. Present the results from step 3 in an easy to understand and concise form.
5. Summarise the main outcomes and conclusions from step 4
6. Reflect on the outcomes from step 5 including:
    a. what limitations apply to the outcomes of your study(due to tools, data, approach, etc)
    b. how you would overcome these limitations if you had more time
    c. a response to your original design idea from step 1

3.2 Submission Instructions – Part 3

You must submit:

- A report of up to 6 pages that documents the key elements from steps 1 to 6 above
- 1 simulation file that represents the key elements of your study.

Present your work clearly and concisely and remember to use the assessment criteria to guide you. High marks will come from correct and insightful application of the skills learned in this unit NOT lengthy submissions.

Submit your work to “Part 3 – Further Study” in the “Assessments” area of the Blackboard site.

3.3 Example Studies:

	Example 1	Example 2	Example 3	Example 4
Machines and Drivetrain	Impact of the counterweight mass on efficiency and/or total energy usage.	Closed loop control of the machine to improve the current response in simulation	Impact of changing machine pole number alongside gear ratio of efficiency and/or DC Link suitability	Impact of changing operation to a non-salient machine and adopting a maximum torque per amp strategy on efficiency.
Converter	Impact of deadtime on harmonics and motor losses	Impact of DC-link ripple on output harmonics and motor losses	Impact of modulation schemes (not just those covered in activity sheet 7) on harmonics & motor losses	